Econ 101A
Section 13: Solutions

Clotaire Boyer*

Fall 2026

Last section we found prices and quantities. Today we ask whether the outcome is any good. The yardstick is surplus: what buyers would have paid on top of what they pay, and what sellers receive on top of their costs. With it we can put a number on the cost of a tax, the gains from trade, and the damage from rent control. Then we let firms enter, which pushes profits to zero. We end with a first look at monopoly. This is the second half of Lecture 9 and the start of Lecture 10.

We reuse one market all day: 10 firms with cost $c(y) = 4 + y^2$ (the 4 is avoided by shutting down), so $Y^S = 5p$ for $p \geq 4$ and 0 below; and 5 consumers, so $X^D = 100 - 5p$. Without a tax, $p^* = 10$, $Q^* = 50$, and each firm earns 21.

1 Producer Surplus

Producer surplus is profit, $\pi(p, y_0) = p\, y_0 - c(y_0)$. Two pictures:

Rectangle. $\pi = y_0 \big(p - \mathrm{AC}(y_0)\big)$: quantity times the margin over average cost.

Area above MC. Since $c(y_0) = c(0) + \int_0^{y_0} c'(y)\, dy$,

$$\pi = \int_0^{y_0} \big(p - c'(y)\big)\, dy - c(0).$$

Profit is the area between the price and marginal cost, minus the fixed cost.

A third picture is often the easiest. By the envelope theorem $d\pi/dp = y(p)$, and profit is zero at the shutdown price $\underline{p}$. So $\pi(p) = \int_{\underline{p}}^{p} y(s)\, ds$: the area to the left of the supply curve, between $\underline{p}$ and p.

*These notes are a consolidation of the notes of Dan Acland, Ivan Balbuzanov, Mariana Carrera, Justin Gallagher, Anne Karing, Matt Leister, Sam Leone, Nicholas Li, Pablo Munoz, Yassine Sbai Sassi, Jon Schellenberg, Katalin Springel, Jonas Tungodden, David Wu, Sam Wang, and Matteo Saccarola.

2 Consumer Surplus

When the price goes from p_0 to p_1, how much money would make the consumer as well off as before? That is $e(p_1, u) - e(p_0, u)$. By the envelope theorem $\partial e/\partial p = h(p, u)$, the Hicksian demand. So

$$e(p_1, u) - e(p_0, u) = \int_{p_0}^{p_1} h(p, u)\, dp,$$

the area to the left of the *Hicksian* demand curve. In practice we use the area under the ordinary (Marshallian) demand curve. The two coincide when there are no income effects, and are close when the good is a small share of the budget.

3 Total Surplus and Deadweight Loss

Total surplus is consumer surplus plus producer surplus plus any tax revenue. In a competitive market without distortions it is as large as it can be. A tax, a tariff, or a price ceiling makes some trades that are worth more than they cost not happen. The surplus lost on those trades is the deadweight loss.

4 The Long Run

In the long run firms can enter. They enter as long as profit is positive, that is, as long as $p > \text{AC}$. Entry shifts industry supply right and pushes the price down until $p = \min \text{AC}$. At that point $\text{MC} = \text{AC}$ and each firm earns zero. If each firm's cost does not depend on how many firms there are (a *constant cost industry*), the long-run supply curve is flat at $\min \text{AC}$.

5 Exercises

Exercise 1 (producer surplus three ways) A firm has $c(y) = 4 + y^2$ for $y > 0$, $c(0) = 0$. The price is $p = 10$.

(a) Find output and profit.

(b) Compute profit as a rectangle, $y_0\,(p - \text{AC})$.

(c) Compute profit as the area between p and MC, minus the setup cost.

(d) Compute profit as the area to the left of the supply curve, between the shutdown price and p. Why does this work? Use it to get industry producer surplus with 10 firms.

Solution.

(a) $p = \text{MC} = 2y$ gives $y_0 = 5$. Profit: $10 \cdot 5 - (4 + 25) = 50 - 29 = 21$.

(b) $\mathrm{AC}(5) = \dfrac{4+25}{5} = \dfrac{29}{5} = 5.8$. So $y_0(p - \mathrm{AC}) = 5\,(10 - 5.8) = 5 \cdot 4.2 = 21$.

(c) Treat the 4 as the fixed cost $c(0)$ in the lecture formula:

$$\int_0^5 (10 - 2y)\,dy - 4 = \Big[10y - y^2\Big]_0^5 - 4 = (50 - 25) - 4 = 21.$$

(d) From Section 12, $y(s) = s/2$ for $s \geq 4$ and the shutdown price is 4. Then

$$\int_4^{10} \frac{s}{2}\,ds = \Big[\frac{s^2}{4}\Big]_4^{10} = \frac{100}{4} - \frac{16}{4} = 25 - 4 = 21.$$

Why it works: by the envelope theorem, $\pi'(p) = y(p)$. At the shutdown price profit is zero, $\pi(4) = 4 \cdot 2 - (4+4) = 0$. So

$$\pi(10) = \pi(4) + \int_4^{10} \pi'(s)\,ds = 0 + \int_4^{10} y(s)\,ds.$$

The setup cost is not forgotten: it is what sets the starting point at 4 instead of 0.

Industry: $\int_4^{10} 5s\,ds = \Big[\frac{5s^2}{2}\Big]_4^{10} = 250 - 40 = 210 = 10 \cdot 21$.

Exercise 2 (when is the area under demand the right measure?)

(a) A consumer has quasilinear utility $u(x, m) = 20x - \frac{1}{2}x^2 + m$, where m is money spent on everything else. Find the Marshallian demand, the Hicksian demand and the expenditure function. Check that $\partial e/\partial p = h$. Find the welfare loss when p rises from 10 to 12, and the consumer surplus at $p = 10$.

(b) A consumer has $u(x, y) = x^{1/2}y^{1/2}$, income $M = 100$, $p_y = 1$. The price of x rises from 1 to 4. Find the Hicksian demand and expenditure function. How much money would the consumer need to be as well off as before? Compare with the area under the Marshallian demand curve.

Solution.

(a) Marshallian: maximize $20x - \frac{1}{2}x^2 + M - px$. FOC: $20 - x - p = 0$, so $x(p) = 20 - p$. Income does not appear.

Hicksian: minimize $px + m$ subject to $20x - \frac{1}{2}x^2 + m = u$. Use the constraint to write $m = u - 20x + \frac{1}{2}x^2$, so we minimize

$$px + u - 20x + \tfrac{1}{2}x^2.$$

FOC: $p - 20 + x = 0$, so $h(p, u) = 20 - p$. Same as Marshallian: no income effects.

Expenditure function: plug $x = 20 - p$ back in.

$$e(p,u) = p(20-p) + u - 20(20-p) + \tfrac{1}{2}(20-p)^2 = u - (20-p)^2 + \tfrac{1}{2}(20-p)^2 = u - \tfrac{1}{2}(20-p)^2.$$

Check: $\partial e/\partial p = -\frac{1}{2}\cdot 2(20-p)\cdot(-1) = 20-p = h$. Good.

Loss from $p=10$ to $p=12$:

$$e(12,u) - e(10,u) = \Big(u - \tfrac{1}{2}\cdot 8^2\Big) - \Big(u - \tfrac{1}{2}\cdot 10^2\Big) = -32 + 50 = 18.$$

Area: $\int_{10}^{12}(20-p)\,dp = \big[20p - \frac{p^2}{2}\big]_{10}^{12} = (240-72) - (200-50) = 168 - 150 = 18$. Same.

Consumer surplus at $p=10$: $\int_{10}^{20}(20-p)\,dp = \big[20p - \frac{p^2}{2}\big]_{10}^{20} = 200 - 150 = 50$ per consumer, so 250 for the five consumers in our market. With quasilinear utility the area under demand is exact.

(b) Hicksian demand. Minimize $p_x x + p_y y$ subject to $x^{1/2}y^{1/2} = u$. Tangency: MRS $= y/x = p_x/p_y$, so $y = \frac{p_x}{p_y}x$. In the constraint:

$$x^{1/2}\Big(\frac{p_x}{p_y}x\Big)^{1/2} = u \implies x\Big(\frac{p_x}{p_y}\Big)^{1/2} = u \implies h_x = u\Big(\frac{p_y}{p_x}\Big)^{1/2}, \quad h_y = u\Big(\frac{p_x}{p_y}\Big)^{1/2}.$$

Expenditure:

$$e = p_x h_x + p_y h_y = u\sqrt{p_x p_y} + u\sqrt{p_x p_y} = 2u\sqrt{p_x p_y}.$$

Initial utility. Marshallian demand is $x = M/(2p_x) = 50$ and $y = M/(2p_y) = 50$, so $u_0 = \sqrt{50\cdot 50} = 50$. Check: $e(1,1,50) = 2\cdot 50\cdot 1 = 100 = M$.

Money needed at the new price:

$$e(4,1,50) - e(1,1,50) = 2\cdot 50\cdot\sqrt{4} - 100 = 200 - 100 = 100.$$

The same number as an area under Hicksian demand:

$$\int_1^4 50\,p^{-1/2}\,dp = 50\Big[2\sqrt{p}\Big]_1^4 = 50\,(4-2) = 100.$$

Area under Marshallian demand $x = 50/p$:

$$\int_1^4 \frac{50}{p}\,dp = 50\big[\ln p\big]_1^4 = 50\ln 4 \approx 69.3.$$

The Marshallian area is too small. As the price rises, the consumer gets poorer and buys even less of x (a normal good). Marshallian demand falls faster than Hicksian demand, so the area under it is smaller. The gap is big here because x is half the budget and the price quadruples. (If we used Hicksian demand at the *new* utility, $u_1 = \sqrt{12.5\cdot 50} = 25$, we would get $2\cdot 25\cdot(2-1) = 50$. The Marshallian area sits between the two.)

Exercise 3 (the cost of a tax) Use the market above. Consumers have the quasilinear utility of Exercise 2(a), so the area under demand is exact.

(a) Compute consumer surplus, producer surplus and total surplus without a tax.

(b) A tax of $t = 4$ per unit is levied. From Section 12, buyers pay 12, sellers keep 8, $Q = 40$. Compute consumer surplus, producer surplus, tax revenue, total surplus and the deadweight loss. Draw the picture.

(c) Check the deadweight loss with a triangle. Who loses what?

(d) Double the tax to $t = 8$. What happens to revenue and to the deadweight loss?

(e) Suppose instead supply were vertical. What would the deadweight loss of a tax be?

Solution.

(a) Consumer surplus: triangle between demand and $p = 10$, base 50, height $20 - 10 = 10$: $\frac{1}{2} \cdot 50 \cdot 10 = 250$. Producer surplus: 210 (Exercise 1). Total: 460.

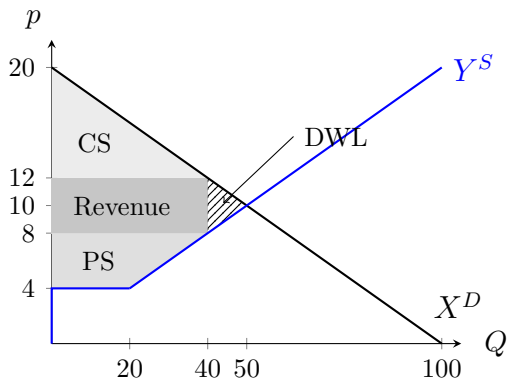
(b) Consumer surplus: base 40, height $20 - 12 = 8$: $\frac{1}{2} \cdot 40 \cdot 8 = 160$.

Producer surplus at the price sellers keep, 8:

$$\int_4^8 5s \, ds = \left[\frac{5s^2}{2}\right]_4^8 = 160 - 40 = 120.$$

Check per firm: each produces 4 and earns $8 \cdot 4 - (4 + 16) = 12$, and $10 \cdot 12 = 120$.

Revenue: $4 \cdot 40 = 160$. Total: $160 + 120 + 160 = 440$. Deadweight loss: $460 - 440 = 20$.



(c) The lost trades are units 40 to 50. On each, the buyer's value exceeds the seller's cost. The triangle has base $50 - 40 = 10$ and height $12 - 8 = 4$: $\frac{1}{2} \cdot 10 \cdot 4 = 20$.

Consumers lose $250 - 160 = 90$. Producers lose $210 - 120 = 90$. The government gets 160. Net: $90 + 90 - 160 = 20$. Most of what buyers and sellers lose is a transfer. Only the triangle is waste.

(d) With $t = 8$: buyers pay $10 + 8/2 = 14$, sellers keep 6, $Q = 50 - 5 \cdot 8/2 = 30$. Sellers' price is still above 4, so firms stay open.

Revenue: $8 \cdot 30 = 240$. Deadweight loss: $\frac{1}{2} \cdot (50 - 30) \cdot 8 = 80$.

Check: CS $= \frac{1}{2} \cdot 30 \cdot 6 = 90$, PS $= \int_4^6 5s\, ds = 90 - 40 = 50$, total $= 90 + 50 + 240 = 380$, and $460 - 380 = 80$.

Doubling the tax multiplies revenue by 1.5 but the deadweight loss by 4. Both the base and the height of the triangle double, so the loss grows with t^2. This is why economists prefer many small taxes to a few big ones.

(e) Zero. If supply is vertical, quantity does not change with the tax. No trade is lost; sellers just keep less (Section 12). The tax is a pure transfer.

Exercise 4 (opening to trade) The same domestic market (10 firms, short run) opens to trade. The good can be bought on the world market at price p_T, in any quantity.

(a) Let $p_T = 6$. Find domestic production, consumption and imports. Compute CS, PS and total surplus. Who wins, who loses, and what is the net gain? Draw the picture.

(b) The government adds a tariff of 2 per imported unit. Redo (a). Where is the deadweight loss?

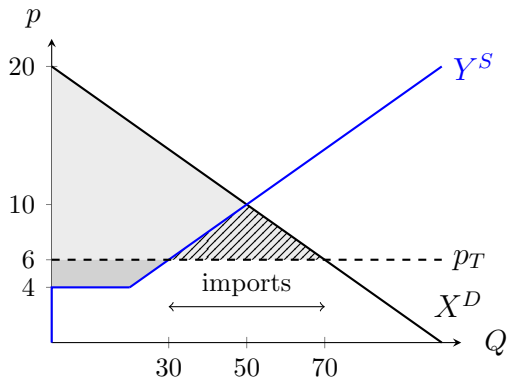
(c) Let $p_T = 3$ instead. What happens to the domestic industry?

Solution.

(a) The domestic price falls to 6. Domestic production $Y^S(6) = 30$. Consumption $X^D(6) = 100 - 30 = 70$. Imports $70 - 30 = 40$.

CS: $\frac{1}{2} \cdot 70 \cdot (20 - 6) = \frac{1}{2} \cdot 70 \cdot 14 = 490$. PS: $\int_4^6 5s\, ds = \left[\frac{5s^2}{2}\right]_4^6 = 90 - 40 = 50$. Total: 540.

Compared with no trade (CS 250, PS 210, total 460): consumers gain 240, producers lose 160, the country gains 80. The gain is the triangle between the curves above $p = 6$: base $70 - 30 = 40$, height $10 - 6 = 4$, area $\frac{1}{2} \cdot 40 \cdot 4 = 80$.



Light gray: CS. Dark gray: PS. Hatched: gain from trade.

(b) Domestic price becomes $6 + 2 = 8$. Production $5 \cdot 8 = 40$, consumption $100 - 40 = 60$, imports 20. Tariff revenue $2 \cdot 20 = 40$.

CS $= \frac{1}{2} \cdot 60 \cdot 12 = 360$. PS $= \int_4^8 5s\, ds = 160 - 40 = 120$. Total $= 360 + 120 + 40 = 520$. Loss relative to free trade: $540 - 520 = 20$.

The loss is two triangles. Production: units 30 to 40 are now made at home at a cost above 6, the world price. Area $\frac{1}{2} \cdot (40 - 30) \cdot 2 = 10$. Consumption: units 60 to 70 are no longer bought though they are worth more than 6. Area $\frac{1}{2} \cdot (70 - 60) \cdot 2 = 10$. Total 20.

(c) At $p = 3$ every firm is below its shutdown price of 4, so domestic production is zero. The industry closes. Consumption $100 - 15 = 85$, all imported. CS $= \frac{1}{2} \cdot 85 \cdot 17 = 722.5$, PS $= 0$. Total surplus is much higher than 460, but every firm closes and its workers lose their jobs, which our model does not count. This is the "difficult trade-off" in the lecture: the winners could compensate the losers, but nothing makes them do it.

Exercise 5 (rent control) In a rental market, demand is $Q^D = 120 - 2p$ and supply is $Q^S = 2p - 40$. The city caps rent at $\bar{p} = 30$.

(a) Find the equilibrium without the cap, and CS, PS.

(b) With the cap, how many units are rented? How many people want one?

(c) *Cost 1.* Assume the units go to the people who value them most. Compute CS, PS and the deadweight loss. Do tenants gain?

(d) *Cost 2.* Now assume the units go at random to people willing to pay at least 30. Compute CS and the total loss. Do tenants as a group still gain?

(e) In the long run, landlords can convert rental units into condos. What does that do to the cost of rent control?

Solution.

(a) $120 - 2p = 2p - 40$, so $4p = 160$, $p^* = 40$, $Q^* = 40$. Inverse curves: demand $p = 60 - Q/2$, supply $p = 20 + Q/2$.

CS $= \frac{1}{2} \cdot 40 \cdot (60 - 40) = 400$. PS $= \frac{1}{2} \cdot 40 \cdot (40 - 20) = 400$. Total 800.

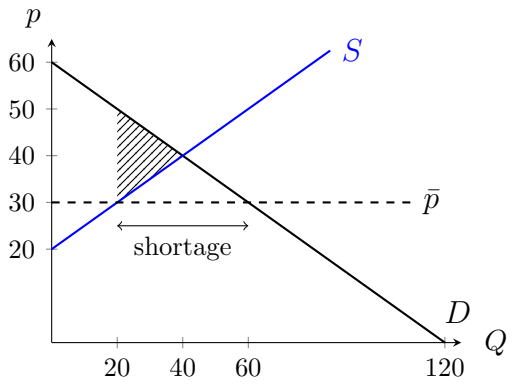
(b) At $p = 30$: $Q^S = 60 - 40 = 20$ units offered and $Q^D = 120 - 60 = 60$ people want one. Shortage of 40.

(c) Only 20 units are rented. With efficient allocation they go to the 20 tenants with the highest values, $60 - Q/2$ for $Q \in [0, 20]$:

$$\text{CS} = \int_0^{20} \Big(60 - \frac{Q}{2} - 30\Big) dQ = \Big[30Q - \frac{Q^2}{4}\Big]_0^{20} = 600 - 100 = 500.$$

$$\text{PS} = \int_0^{20} \Big(30 - 20 - \frac{Q}{2}\Big) dQ = \Big[10Q - \frac{Q^2}{4}\Big]_0^{20} = 200 - 100 = 100.$$

Total 600, so the deadweight loss is 200. Triangle check: at $Q = 20$ demand price is 50 and supply price is 30, so area $\frac{1}{2} \cdot (40 - 20) \cdot (50 - 30) = 200$. Tenants gain $500 - 400 = 100$. Landlords lose $400 - 100 = 300$.



(d) The 60 people willing to pay 30 have values spread evenly from 60 down to 30, so the average is 45. A random 20 of them get a unit. Their total value is $20 \cdot 45 = 900$, and they pay $20 \cdot 30 = 600$. CS = 300. PS is still 100. Total 400.

The loss is now $800 - 400 = 400$: twice Cost 1. The extra 200 is misallocation. The efficient 20 have average value $(60+50)/2 = 55$, so they would get $20 \cdot 55 = 1100$; random assignment gives 900.

Tenants as a group now have 300, less than the 400 they had without the cap. The lucky few gain a lot. Everyone else gains nothing.

(e) It makes supply more elastic. A more elastic supply means more units are lost at the capped price, so a bigger triangle. Diamond, McQuade and Qian (2019, *AER*) study the 1994 expansion of rent control in San Francisco and find that landlords of newly covered buildings cut the supply of rental housing by about 15%, often by converting to condos.

Exercise 6 (free entry) Keep the firm $c(y) = 4 + y^2$ and demand $X^D = 100 - 5p$, but now any number of identical firms can enter.

(a) Find the long-run price, output per firm, total quantity and number of firms.

(b) Demand rises to $X^D = 140 - 5p$. Find the short-run equilibrium with the number of firms from (a), then the new long-run equilibrium.

(c) Back to $X^D = 100 - 5p$. A tax of $t = 2$ per unit is levied. Find the long-run outcome. Who pays the tax? Compare with Section 12, Exercise 5.

Solution.

(a) Entry pushes the price to $\min \mathrm{AC} = 4$ (Section 12, Exercise 1). Each firm produces $y = 2$ and earns zero. Total $Q = 100 - 5 \cdot 4 = 80$. Number of firms $J = 80/2 = 40$. (With 10 firms they earned 21 each, so 30 more firms enter.)

(b) Short run with $J = 40$: $Y^S = 40 \cdot p/2 = 20p$. Then

$$20p = 140 - 5p \implies 25p = 140 \implies p = 5.6.$$

Each firm produces $5.6/2 = 2.8$ and earns $5.6 \cdot 2.8 - (4 + 2.8^2) = 15.68 - 11.84 = 3.84 > 0$. Entry follows.

Long run: price back to 4, $Q = 140 - 20 = 120$, $J = 120/2 = 60$. The demand shift raised the price only for a while. In a constant cost industry the long-run supply curve is flat at 4, and all the adjustment is in the number of firms.

(c) In the long run sellers must keep 4, so buyers pay $4 + 2 = 6$. $Q = 100 - 30 = 70$, each firm still makes 2, so $J = 35$. Buyers pay the whole tax: long-run supply is flat ($\varepsilon_{S,p} \to \infty$). In Section 12, with 10 firms fixed, buyers paid only half. The deadweight loss is $\frac{1}{2} \cdot 2 \cdot (80 - 70) = 10$.

Exercise 7 (preview: monopoly) Now one firm serves the whole market. It produces at constant marginal cost 4 with no fixed cost. Inverse demand is $p = 20 - Q/5$.

(a) Find the monopoly quantity, price and profit.

(b) Check the markup rule $\dfrac{p - \mathrm{MC}}{p} = -\dfrac{1}{\varepsilon_{Q,p}}$.

(c) Compare with the competitive outcome. What is the deadweight loss?

(d) Compare with a tax of $t = 8$ in the competitive industry of Exercise 6. What do you notice?

Solution.

(a) Revenue $R = (20 - Q/5)\,Q = 20Q - Q^2/5$. Marginal revenue $\mathrm{MR} = 20 - 2Q/5$. Set MR = MC:

$$20 - \frac{2Q}{5} = 4 \implies \frac{2Q}{5} = 16 \implies Q^M = 40.$$

Price from the *demand* curve, not the MR curve: $p^M = 20 - 40/5 = 12$. Profit $= (12-4) \cdot 40 = 320$. SOC: $\mathrm{MR}' = -2/5 < 0$.

(b) $\dfrac{p - \mathrm{MC}}{p} = \dfrac{12 - 4}{12} = \dfrac{2}{3}$. Elasticity at $(40, 12)$: $Q = 100 - 5p$, so $\varepsilon_{Q,p} = -5 \cdot \dfrac{12}{40} = -1.5$, and $-1/\varepsilon = 1/1.5 = 2/3$. The rule holds.

(c) Competition: $p = \mathrm{MC} = 4$, $Q = 80$, $\mathrm{CS} = \frac{1}{2} \cdot 80 \cdot 16 = 640$, total 640. Monopoly: $\mathrm{CS} = \frac{1}{2} \cdot 40 \cdot 8 = 160$, profit 320, total 480. Deadweight loss $= 640 - 480 = 160 = \frac{1}{2} \cdot (12-4) \cdot (80-40)$.

(d) A tax of 8 on a flat supply curve at 4 gives buyers' price 12 and $Q = 100 - 60 = 40$. Same price, same quantity, same deadweight loss of 160. Tax revenue $8 \cdot 40 = 320$ equals monopoly profit. A monopolist acts like a tax collector who keeps the revenue.